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1. If we keep the three parts of the network (feature extractor, label classifier, domain classifier) but remove the GRL, the model will no longer learn domain-invariant features. The domain classifier will simply learn to separate MNIST and MNIST-M, and the feature extractor will keep domain-specific information instead of hiding it.
As a result, the model behaves like a naive network: it reaches 99% accuracy on MNIST, but drops to about 51% on MNIST-M, ~~exactly~~ Domain adaptation completly fails without the GRL.

2. The performance on the source dataset may drop because the DANN model is no longer optimizing only for source classification. With the ~~grdi~~ GRL, the feature extractor is also pushed to make the representations domain-invariant so that the domain classifier cannot distinguish source from target samples. This adversarial objective forces the network to remove domain-specific information that is actually useful for classifying source images. As a result, the learned features become more generic and less specialized for the source domain, which can slightly reduce source accuracy.

3. The negative factor in the GRL controls the strength of the adversarial signal. If the factor is too small, the gradient reversal is weak, the model keeps learning source-specific features, and domain adaptation remains poor. If the factor is too large too early, the adversial pressure become becomes unstable and can damage the classifier before it has learned good features. Using a gradually increasing factor avoids instability and allows the model to learn both ~~the~~ discriminative and domain-invariant features, leading to effective domain adaptation.

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4. Pseudo-labeling is ~~a tech~~ to let the model predict labels for the target samples, threat these predictions as " fake labels", and then train on them as if they were real. In this way, the model gradually learns to adapt to the target domain even without true annotations.

The process typically works in two steps:

(1) train a model on the labeled source domain and use it to make predictions on the unlabeled target data.

(2) select ~~condit~~ confident predictions (for example, predictions with high softmax probability) and use them as pseudo-labels to continue training the model together with the labeled sorce data.

By repeatedly refining the pseudo-labels and retraining, the model becomes better at the target domain. The key idea is that the model teaches itself on the target ~~and~~ domain using its own predictions.

In summary, pseudo-labeling allows domain adaptation by turning unlabeled target samples into training data through confident self-generated labels.

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